ETSI TS 104 258 V2.1.1 (2026-10)



TECHNICAL SPECIFICATION

Intelligent Transport Systems (ITS);
Facilities Layer;
Point of Interest Message (POIM) Format;
Release 2

Reference
DTS/ITS-0011014

Keywords
automotive, ITS, safety, service, transport

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° w061004871

Important notice

The present document can be downloaded from the
ETSI Search & Browse Standards application.

The present document may be made available in electronic versions and/or in print. The content of any electronic and/or print versions of the present document shall not be modified without the prior written authorization of ETSI. In case of any existing or perceived difference in contents between such versions and/or in print, the prevailing version of an ETSI deliverable is the one made publicly available in PDF format on ETSI deliver repository.

Users should be aware that the present document may be revised or have its status changed, this information is available in the Milestones listing.

If you find errors in the present document, please send your comments to the relevant service listed under Committee Support Staff.

If you find a security vulnerability in the present document, please report it through our Coordinated Vulnerability Disclosure (CVD) program.

Notice of disclaimer & limitation of liability

The information provided in the present deliverable is directed solely to professionals who have the appropriate degree of experience to understand and interpret its content in accordance with generally accepted engineering or other professional standard and applicable regulations.
No recommendation as to products and services or vendors is made or should be implied.
No representation or warranty is made that this deliverable is technically accurate or sufficient or conforms to any law and/or governmental rule and/or regulation and further, no representation or warranty is made of merchantability or fitness for any particular purpose or against infringement of intellectual property rights.
In no event shall ETSI be held liable for loss of profits or any other incidental or consequential damages.

Any software contained in this deliverable is provided "AS IS" with no warranties, express or implied, including but not limited to, the warranties of merchantability, fitness for a particular purpose and non-infringement of intellectual property rights and ETSI shall not be held liable in any event for any damages whatsoever (including, without limitation, damages for loss of profits, business interruption, loss of information, or any other pecuniary loss) arising out of or related to the use of or inability to use the software.

Copyright Notification

No part of this document may be reproduced in any form, by any means and in any media, without the prior written authorization of ETSI and except as expressly permitted below.
By way of exception and when the document is a normative deliverable (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)), ETSI authorizes to reproduce and incorporate into products, services and technical documentation only those extracts (e.g. templates) that are strictly necessary for the technical implementation of the normative deliverable, to ensure compliance with the latter.
Nothing in this notice shall be construed as limiting any mandatory exceptions to copyright provided by applicable law.

© ETSI 2026.
All rights reserved.

Contents

Intellectual Property Rights 4
Foreword 4
Modal verbs terminology 4
Introduction 4
1 Scope 6
2 References 6
2.1 Normative references 6
2.2 Informative references 6
3 Definition of terms, symbols and abbreviations 6
3.1 Terms 6
3.2 Symbols 6
3.3 Abbreviations 7
4 POIM specification 7
4.1 POIM Structure 7
4.1.1 General structure of a POIM 7
4.1.2 ITS PDU header 8
4.1.3 POI Information Block 8
4.1.4 POIM common containers 8
4.1.4.1 POI Block Management Container 8
4.1.4.2 Related POI Info Block Container 8
4.2 POIM format specification 9
Annex A (normative): ASN.1 modules 10
History 11

Intellectual Property Rights

Essential patents

IPRs essential or potentially essential to normative deliverables (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)) may have been declared to ETSI. The declarations pertaining to these essential IPRs, if any, are publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI IPR online database.

Pursuant to the ETSI Directives including the ETSI IPR Policy, no investigation regarding the essentiality of IPRs, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Trademarks

The present document may include trademarks and/or tradenames which are asserted and/or registered by their owners. ETSI claims no ownership of these except for any which are indicated as being the property of ETSI, and conveys no right to use or reproduce any trademark and/or tradename. Mention of those trademarks in the present document does not constitute an endorsement by ETSI of products, services or organizations associated with those trademarks.

DECT™, **PLUGTESTS™**, **UMTS™** and the ETSI logo are trademarks of ETSI registered for the benefit of its Members. **3GPP™**, **LTE™** and **5G™** logo are trademarks of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners. **oneM2M™** logo is a trademark of ETSI registered for the benefit of its Members and of the oneM2M Partners. **GSM**® and the GSM logo are trademarks registered and owned by the GSM Association.

Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Intelligent Transport Systems (ITS).

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

Introduction

The present document specifies the format of the generic Point Of Interest Message (POIM), which is a facilities layer protocol data unit that can be used to disseminate various types of information regarding so called Points-of-Interest (POIs). These POIs are locations that can be useful for and/or of particular interest to road transport users, such as parking facilities, energy (e.g. fuelling and/or charging) facilities, public transport hubs and/or any combination of those.

The POIM represents a class of messages that can be used to instantiate specific messages for specific purposes and for specific POIs. It has a modular structure essentially composed of POI information blocks. Each POI information block can be specified for a specific type of POI. The specification of such blocks is out of the scope of the present document: in fact the present document is intended to be referenced by standards that specify facilities layer services which disseminate information related to specific types of POIs. These standards are expected to specify all aspects related to a facility layer service such functional aspects, dissemination requirements and message formats. Hence it is up to those standards to define the necessary POI information blocks and to define the corresponding message(s) based on the POIM format specified in this present document. Refer to the specification of the Parking Availability Service in ETSI TS 103 916 [i.2] as an example.

Additionally, the present document also specifies the format of common containers that are recommended to be used when defining POI information blocks.

1 Scope

The present document specifies the abstract syntax of the Point Of Interest Message (POIM), and the abstract syntax and semantics of its common containers.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 102 894-2: "Intelligent Transport Systems (ITS); Facilities Layer; Part 2: Common Data Dictionary (CDD); Release 2".

[2] Recommendation ITU-T X.680: "Information technology - Abstract Syntax Notation One (ASN.1): Specification of basic notation".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI TR 103 902: "Intelligent Transport Systems (ITS); ITS Framework; Terms, Symbols and Abbreviations; Release 2".

[i.2] ETSI TS 103 916: "Intelligent Transport Systems (ITS); Parking Availability Service; Release 2".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI TR 103 902 [i.1] apply.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI TR 103 902 [i.1] apply.

4 POIM specification

4.1 POIM Structure

4.1.1 General structure of a POIM

A POIM is a PDU composed of one common ITS PDU header and multiple POIM Blocks, which together constitute the POIM payload. Each POI Information Block can be generated and processed independently from the other, whilst linkage between the POI Information Blocks is possible.

The general structure of a POIM is illustrated in Figure 1.

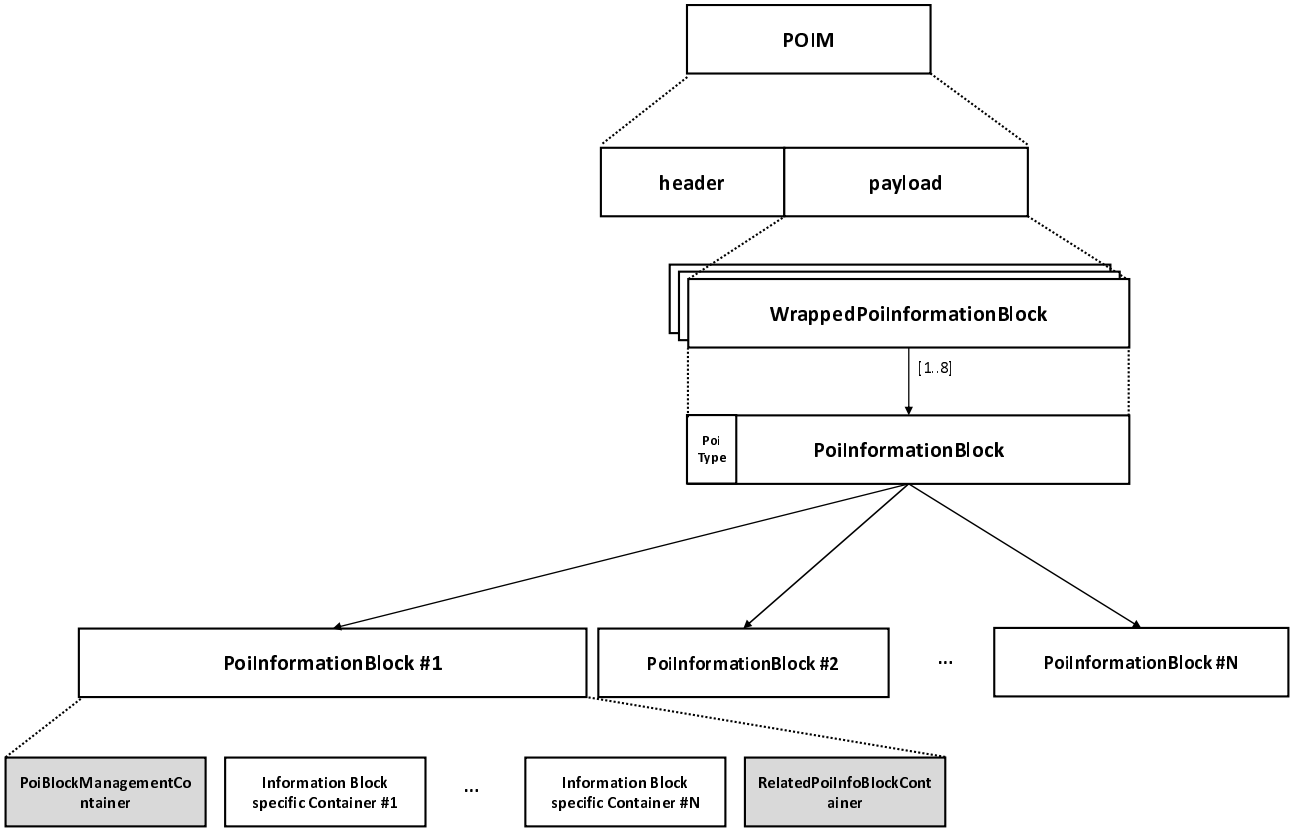


Figure 1: General Structure of a POIM

The component *header* is a common header for facility layer PDUs.

The component *payload* represents the POIM payload and shall consist of a list of POI Information Blocks wrapped in a structure called *WrappedPoiInformationBlocks* that includes the type identifier of the Information Block and the Information Block itself.

POI Information Blocks can be defined for Point-of-Interest specific aspects, i.e. aspects related to a specific location and purpose that are useful or interesting such as: parking availability, energy supply, electric charging, public transport, etc.

The POIM shall contain at least one POI Information Block for a specific type of POI. The message that is instantiated shall be named "POIM-XY" where XY is the abbreviation of the specific type of POI such as for example PA for Parking Availability or EC for Electric Charging. If the POIM contains more than one POI Information Block it can be considered a combined message, e.g. it can be considered both a POIM-PA and a POIM-EC.

4.1.2 ITS PDU header

The ITS PDU header shall be as specified in ETSI TS 102 894-2 [1] and as constrained in Annex A.

4.1.3 POI Information Block

Each POI Information Block shall consist of the following containers:

- A POI Block Management Container which has the same format for all POI Information Blocks, see clause 4.2.
- One or more Information Block specific containers. The specification of those containers is outside the scope of the present document.
- An optional Related POI Info Block Container which has the same format for all POI Information Blocks, see clause 4.2.

Each container is composed of a sequence of components, their component type being either a Data Element (DE) or a Data Frame (DF). A component is either optional or mandatory in the POIM Format. If not specified as optional in the ASN.1 specification of the present document, a component is considered as mandatory.

4.1.4 POIM common containers

4.1.4.1 POI Block Management Container

The POI Block Management Container provides basic information which is necessary to process the other information in the POI Information Block.

The POI Block Management Container shall contain the following components:

1) *serviceProviderId* to uniquely identify the POI information issuer.
2) *blockIdentificationNumber* to uniquely identify the POI Information Block.
3) *timestamp* to identify the time at which the POI Block was generated.

The POI Block Management Container may contain the components *linkedIvims* or *linkedMapems* to link the POI Information Block with IVIMs and/or MAPEMs.

4.1.4.2 Related POI Info Block Container

The Related POI Info Block Container provides information related to the POI Information Block which is contained in other Information Blocks. The Related POI Info Block Container shall contain a list of DF *RelatedPoiInfoBlock* pointing to other Information Blocks as applicable.

RelatedPoiInfoBlock shall contain the components *RelatedPoiInfoBlock*, *serviceProviderId* and *blockIdentificationNumber*.

RelatedPoiInfoBlock may contain the component *timestamp* of the related Poi Info Block to refer to one specific update of the block.

4.2 POIM format specification

The abstract syntax and semantics of the POIM and the common containers shall be as specified in ASN.1 in Annex A of the present document.

DEs and DFs that are not defined in the present document shall be imported from the common data dictionary ETSI TS 102 894-2 [1] as specified in Annex A.

The specification of encoding rules is outside of the scope of the present document.

Annex A (normative): ASN.1 modules

This clause provides the normative ASN.1 modules according to Recommendation ITU-T X.680/ [2] containing the syntactical specification of the POIM PDU, its containers, and the data frames and data elements defined in the present document.

The POIM-PDU-Description module is identified by the Object Identifier {itu-t (0) identified-organization (4) etsi (0) itsDomain (5) wg1 (1) poim (104258) poim (0) major-version-1 (1) minor-version-1 (1)}. The module can be downloaded as a file as indicated in Table A.1. The associated SHA-256 cryptographic hash digest of the referenced file offers a mean to verify the integrity of that file.

The semantical specification of the POIM components is contained in the same module in the form of ASN.1 comments and is also available in readable format as indicated in Table A.1.

Table A.1: POIM PDU Description ASN.1 module information

Module Name	POIM-PDU-Description
OID	{itu-t (0) identified-organization (4) etsi (0) itsDomain (5) wg1 (1) poim (104258) poimPdu (0) major-version-1 (1) minor-version-1 (1)}
Link	https://forge.etsi.org/rep/ITS/asn1/poim_ts104258/-/raw/v2.1.1/POIM-PDU-Description.asn
SHA-256 hash	df4992bd554a77b703e242f47f298421845f7a9362a899c8c0bc13c8b07ed66f
Readable format	https://forge.etsi.org/rep/ITS/asn1/poim_ts104258/-/blob/v2.1.1/docs/POIM-PDU-Description.md

The POIM-CommonContainers module is identified by the Object Identifier {itu-t (0) identified-organization (4) etsi (0) itsDomain (5) wg1 (1) poim (104258) common(99) major-version-1 (1) minor-version-1 (1)}. The module can be downloaded as a file as indicated in Table A.2. The associated SHA-256 cryptographic hash digest of the referenced file offers a means to verify the integrity of that file. The semantical specification of the common containers components is contained in the same module in the form of ASN.1 comments and is also available in readable format as indicated in Table A.2.

Table A.2: POIM Common Containers ASN.1 module information

Module Name	POIM-CommonContainers
OID	{itu-t (0) identified-organization (4) etsi (0) itsDomain (5) wg1 (1) poim (104258) common (99) major-version-1 (1) minor-version-1 (1)}
Link	https://forge.etsi.org/rep/ITS/asn1/poim_ts104258/-/raw/v2.1.1/POIM-CommonContainers.asn
SHA-256 hash	26093e20f6f980fb7781d51b8db2cd383684d63fbd9deb80560a342644dd5d5c
Readable format	https://forge.etsi.org/rep/ITS/asn1/poim_ts104258/-/blob/v2.1.1/docs/POIM-CommonContainers.md

The POIM-ExternalModule is a template intended to be completed by the user of the standard. It provides a template information object set that is imported by the POIM-PDU-Description module and also includes a "commented-out" practical example for POIM-PA according to [i.2]. The template module can be downloaded as a file as indicated in Table A.3. The semantical specification of the data types is contained in the same module in the form of ASN.1 comments and is also available in readable format as indicated in Table A.3.

Table A.3: POIM External Module ASN.1 module information

Module Name	POIM-ExternalModule
OID	n.a.
Link	https://forge.etsi.org/rep/ITS/asn1/poim_ts104258/-/raw/v2.1.1/POIM-ExternalModule.asn
SHA-256 hash	n.a.
Readable format	https://forge.etsi.org/rep/ITS/asn1/poim_ts104258/-/blob/v2.1.1/docs/POIM-ExternalModule.md

History

Version	Date	Status
V2.1.1	October 2026	Publication